OLA

END-TO-END DATA ANALYSIS

OLA Rides Analytics

July 2024 · Bengaluru
103,024 bookings · 50 areas · 7 vehicle types



Prepared for executive stakeholders

Cleaning · SQL · EDA · Power BI · Streamlit

01 · THE PROBLEM

Nearly 4 in 10 bookings never become a paid ride

62.1%

booking success rate

63,967 completed of 103,024 bookings

The question is not "how many rides?"

It is **where is revenue leaking**, and what is the cheapest lever to stop it?

The four business use cases

1 · Peak demand & driver allocation
When is demand concentrated?

2 · Customer behaviour & marketing
Who are the high-value riders?

3 · Pricing patterns & surge
Does fare track distance?

4 · Anomaly & fraud detection
Zero-distance fares, extreme rates, duplicates

02 · APPROACH

A six-stage pipeline, from raw file to decision

1
Clean
Profile, parse, standardise, engineer 11 features

2
SQL
10 business questions, validated against the CSV

3
EDA
Distributions, trends, correlations, anomalies

4
Power BI
5-page dashboard, 40 DAX measures

5
Streamlit
Natural-language Q&A over SQL

6
Insights
Prioritised, revenue-linked recommendations

Deliverables produced

- ✅ Cleaned CSV (103,024 × 37)
- ✅ SQLite database + 3 indexes
- ✅ queries.sql + SQL_RESULTS.md
- ✅ eda.py + 10 saved charts
- ✅ Power BI guide + theme + DAX
- ✅ 5-page dashboard mockup
- ✅ Streamlit app (5 tabs)
- ✅ Report + business insights

03 · DATASET OVERVIEW

103,024 bookings across 50 Bengaluru areas

RAW FILE

OLA_DataSet.xlsx

Sheet: July

103,024 rows × 20 columns

Period: 1–31 July 2024

Grain

One row per booking attempt — completed, cancelled or unassigned.

7	50	4	4
vehicle types	pickup areas	booking statuses	payment methods

Column groups

Identity

Booking_ID · Customer_ID

Time

Date · Time

Trip

Vehicle_Type · Pickup · Drop Ride_Distance · Booking_Value

Outcome

Booking_Status · Payment_Method Incomplete_Rides + reason

Quality

V_TAT · C_TAT Driver_Ratings · Customer_Rating

04 · DATA CLEANING & QUALITY

Completeness 75.2% → 94.8%, zero rows lost

Before → After scorecard

Metric	Before	After
Rows	103,024	103,024
Columns	20	**37**
Completeness	75.22%	**94.77%**
Duplicate rows	0	0
Unparseable dates	0	0

No fabrication

39,057 non-successful bookings keep NULL ratings/TAT — flagged, never imputed.

11 derived features

Booking_Hour · Booking_DayOfWeek · Is_Weekend · Time_of_Day · Distance_Bucket · Value_Bucket · Fare_per_km · Customer_Cancel_Category · Driver_Cancel_Category · Incomplete_Reason_Category · Zero_Distance_Flag

Cleaning steps (logged)

Date+Time → single timestamp · whitespace trimmed · **eBike → E-Bike**

Structural blanks → "Not Applicable" · IQR outlier flags (8,468 fares)

De-duplicated on Booking_ID · 3 SQLite indexes built

05 · KEY KPIs

₹35.08M revenue from 63,967 completed rides

103,024	62.1%	₹35.08M	₹548
Total bookings	Success rate	Total revenue	Avg booking value

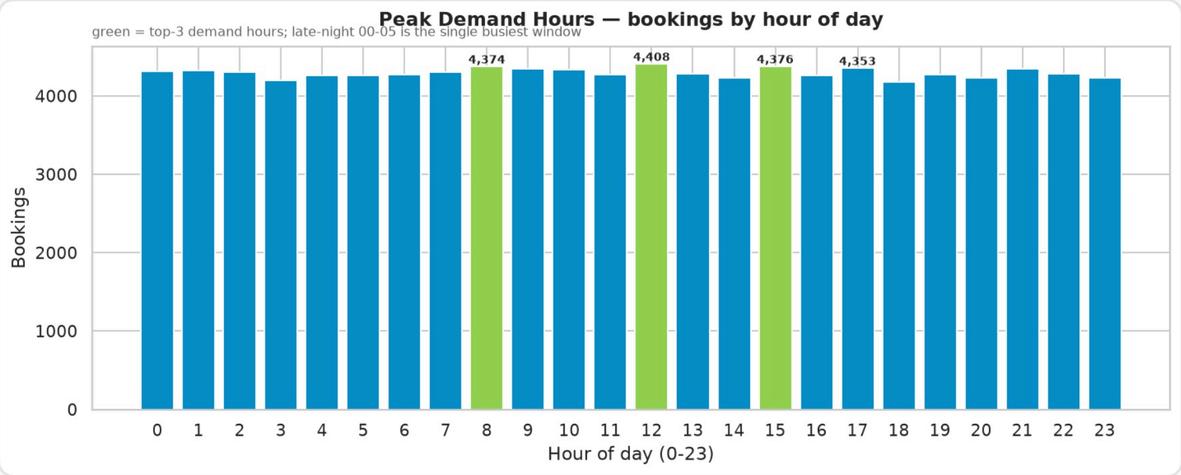
22.85 km	4.00	8.41%	94,544
Avg ride distance	Avg driver rating	Repeat-customer rate	Distinct customers

Booking status mix — where the 37.9% goes



06 · EDA HIGHLIGHTS

Demand is flat; fare is not distance-driven



Hourly volume spans only 4,176–4,408 — a 5.6% spread. No rush hour to exploit.

THE STRUCTURAL FINDING

r = 0.002

Fare ↔ distance correlation. A 5 km and a 45 km ride cost the same on average.

Other EDA findings

- Late night 00–05 is the **largest** demand block (20.8%)
- Auto averages **10.0 km** vs ~25 km for all others
- Fare spread across vehicle types is only **3%**
- Driver ↔ customer rating correlation **r = −0.003**

Anomaly & fraud checks

✅ 0 duplicate Booking_IDs

✅ 0 same-customer-same-time

✅ 0 ratings outside 3.0–5.0

⚠️ 3,591 rides > ₹200/km

⚠️ 889 rides > ₹500/km

07 · THE 10 SQL QUESTIONS

Ten questions, ten validated answers

#	Business question	Answer	Validation
Q1	All successful bookings	**63,967 bookings**	✅ matches scorecard
Q2	Avg ride distance per vehicle type	**Auto 10.0 km · others ~25 km**	✅ sums to 63,967
Q3	Customer-cancelled rides	**10,499**	✅ 10.2% of total
Q4	Top 5 customers by bookings	**CID954071 — 5 bookings**	✅ max = 5
Q5	Driver cancels — personal & car	**6,542 rides**	✅ 35.5% of driver cancels
Q6	Min/max driver rating — Prime Sedan	**3.0 / 5.0 (avg 3.99)**	✅ 9,379 rated rides
Q7	All UPI rides	**25,881 rides**	✅ 40.5% of success
Q8	Avg customer rating per vehicle	**3.99–4.01 (all 7 types)**	✅ 63,967 rated
Q9	Total value of successful rides	**₹35,080,467**	✅ exact match
Q10	All incomplete rides + reason	**3,926 rides**	✅ reasons sum to 3,926

Full SQL, result tables and independent cross-checks: **sql/queries.sql** · **sql/SQL_RESULTS.md**

08 · INSIGHTS BY SEGMENT

Vehicle, payment and cancellation patterns

Avg distance by vehicle

E-Bike	**25.2 km**
Prime Plus	**25.0 km**
Prime Sedan	**25.0 km**
Mini	**25.0 km**
Bike	**24.9 km**
Prime SUV	**24.9 km**
Auto	**10.0 km**

Payment mix (success)

Cash	**35,022 · 54.7%**
UPI	**25,881 · 40.5%**
Credit Card	**2,435 · 3.8%**
Debit Card	**629 · 1.0%**

Cash + UPI = 95.2% · cards only 4.8%

Driver cancel reasons

Personal & Car	**6,542**
Customer related	**5,413**
Customer sick	**3,654**
Capacity violation	**2,825**

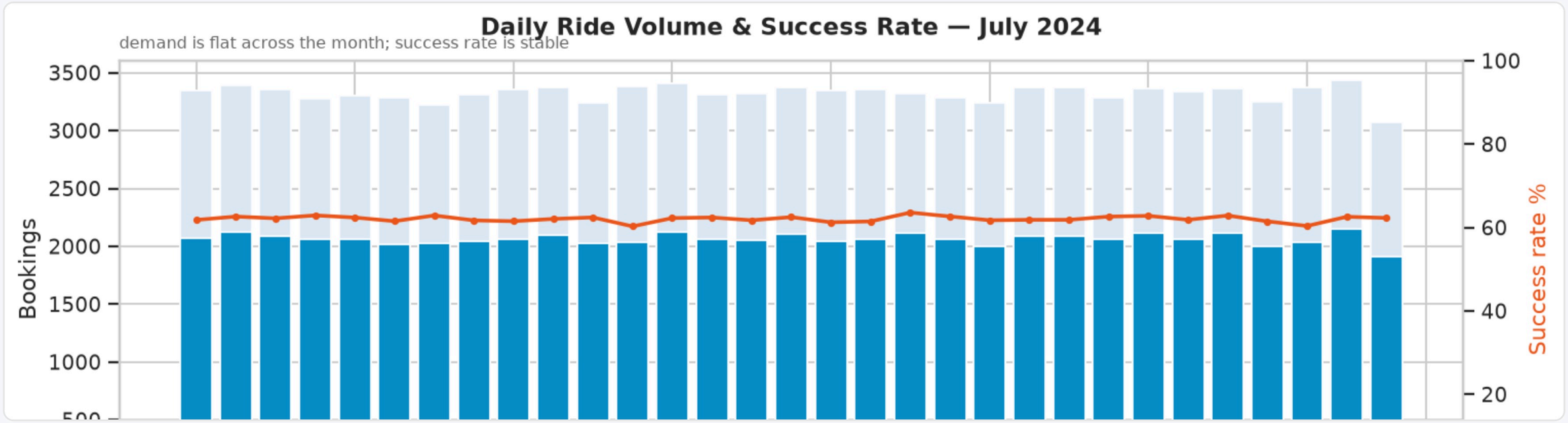
18,434 driver cancels = biggest leak

Customer cancel reasons — driver behaviour dominates

Driver not moving	Driver asked to cancel	Change of plans	AC not working	Wrong address
3,175	**2,670**	**2,081**	**1,568**	**1,005**

09 · TRENDS

Stable volume, stable quality — no drift



3,323
avg daily bookings

3,072–3,432
daily range (31 days)

62.09%
avg success rate

±0.71
success-rate std dev

Read: a 0.71-point standard deviation on a 62% success rate means the operation is in statistical control. The problem is structural, not a bad week — so the fix must be structural too.

10 · POWER BI DASHBOARD

Five pages, one story per page

1 · Overall

Ride Volume Over Time

Booking Status Breakdown

4 KPI cards

2 · Vehicle Type

Top 5 by Ride Distance

Avg Distance by Vehicle

Performance matrix

3 · Revenue

Revenue by Payment Method

Top 5 Customers by Value

Distance per Day

4 · Cancellation

Reasons (Customer)

Reasons (Driver)

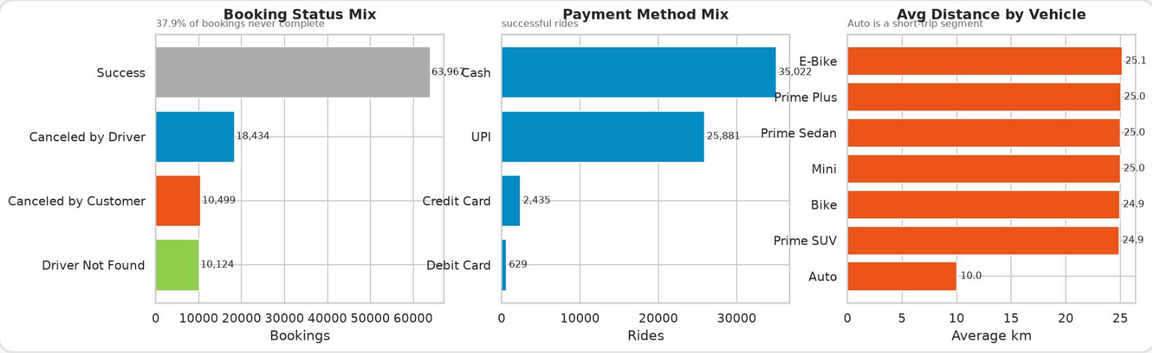
Revenue lost

5 · Ratings

Driver Distribution

Customer Distribution

Customer vs Driver



BUILD KIT INCLUDED

- ✅ Step-by-step build guide
- ✅ theme.json (Ola palette + Poppins)
- ✅ ~40 copy-paste DAX measures
- ✅ Static 5-page HTML mockup
- ✅ .pbit template spec + acceptance test

Slicers synced across all 5 pages: **Date · Vehicle Type · Payment Method · Booking Status**

11 · STREAMLIT APP

Ask a question in English, get SQL and a chart

USER TYPES

```
Which vehicle type earns the most revenue?
```

↓ GENERATED SQL

```
SELECT Vehicle_Type,
  SUM(Booking_Value) AS revenue
FROM bookings
WHERE Booking_Status='Success'
GROUP BY Vehicle_Type
ORDER BY revenue DESC;
```

Five tabs

- 💬 **Ask the Data** — NL → SQL → table → chart
- 🧮 **10 SQL Questions** — one-click run
- 🔎 **Data Explorer** — filters + search
- 📊 **Power BI** — iframe embed + fallback
- ✅ **Data Quality** — scorecard + outliers

Two engines: LLM (if an API key is set) with a robust rule-based fallback covering totals, averages, counts, group-bys, top-N and trends. Read-only SELECT only.

Verified

- ✅ App boots, HTTP 200
- ✅ All 5 tabs render
- ✅ 10/10 queries execute
- ✅ 9 example questions
- ✅ CSV download on every result

12 · RECOMMENDATIONS

Fix supply reliability first — it is worth ₹5M/month

PRIORITY 1

Cut driver cancellations

6,542 "personal & car" cancels. Vehicle-health checks + a reliability score.

+₹5.0M/mo

PRIORITY 2

Resolve the pricing question

Fare ↔ distance r = 0.002. Confirm whether fare is genuine before any surge work.

Data governance

PRIORITY 3

Grow repeat riders

8.41% repeat rate. Target 12% with a ride-3 incentive on the existing cohort.

+3,400 riders

PRIORITY 4

Shift payments to digital

Cards only 4.8%. Cashless incentives cut settlement cost and enable instant refunds.

95.2% → 98%

Also recommended

- **Late-night incentive** — 00–05 is the largest demand block (20.8%)
- **Fare-per-km audit queue** — 889 rides above ₹500/km
- **Stop flagging zero-distance** — 37.9% false positives

The one-line summary: the operation is stable and the customers are satisfied — the money is being lost to driver-side cancellations, and that is a fixable, measurable, high-return problem.

THANK YOU

Questions & next steps

Full deliverables: cleaned data · SQL · EDA · Power BI kit · Streamlit app · report



103,024 bookings

₹35.08M revenue

62.1% success